

Lab 1: Film Ranking (The Basics)

```
1 WITH ranked_film AS (  
2 SELECT  
3 c.name AS category_name,  
4 f.title,  
5 f.length,  
6 DENSE_RANK() OVER (  
7 PARTITION BY c.name  
8 ORDER BY f.length DESC  
9 ) AS length_rank  
10 FROM film f  
11 JOIN film_CATEGORY FC  
12 ON f.film_id = fc.film_ID  
13 JOIN category c  
14 ON fc.category_id = c.category_id  
15 )  
16 SELECT  
17 category_name,  
18 title,  
19 length,  
20 length_rank  
21 FROM ranked_film  
22 WHERE length_rank <= 3  
23 ORDER BY category_name, length_rank, title;
```

Data Output Messages Notifications

Showing rows: 1 to 64 Page No: 1 of 1

	category_name character varying (25)	title character varying (255)	length smallint	length_rank bigint
1	Action	Darn Forrester	185	1
2	Action	Worst Banger	185	1

Total rows: 64 Query complete 00:00:00.093 CRLF Ln 23, Col 43

	category_name character varying (25)	title character varying (255)	length smallint	length_rank bigint
3	Action	Casualties Encino	179	2
4	Action	Quest Mussolini	177	3
5	Animation	Gangs Pride	185	1
6	Animation	Pond Seattle	185	1
7	Animation	Sons Interview	184	2
8	Animation	Theory Mermaid	184	2
9	Animation	Lawless Vision	181	3
10	Children	Fury Murder	178	1
11	Children	Wrong Behavior	178	1
12	Children	Empire Malkovich	177	2
13	Children	Model Fish	175	3
14	Classics	Conspiracy Spirit	184	1
15	Classics	Dracula Crystal	176	2
16	Classics	Lights Deer	174	3
17	Comedy	Control Anthem	185	1
18	Comedy	Saturn Name	182	2
19	Comedy	Searchers Wait	182	2
20	Comedy	Caper Motions	176	3
21	Comedy	Sweden Shining	176	3
22	Documentary	Wife Turn	183	1
23	Documentary	Young Language	183	1
Total rows: 64		Query complete 00:00:00.093		

Lab 2: Growth Analysis (Time Series)

Query Query History

```
25 WITH monthly_revenue AS (  
26 SELECT  
27 DATE_TRUNC('month', payment_date) AS month,  
28 SUM(amount) AS total_revenue  
29 FROM payment  
30 GROUP BY DATE_TRUNC('month', payment_date)  
31 ),  
32 revenue_with_previous AS (  
33 SELECT  
34 month,  
35 total_revenue,  
36 LAG(total_revenue) OVER (  
37 ORDER BY month  
38 ) AS previous_revenue  
39 FROM monthly_revenue  
40 )  
41 SELECT  
42 month,  
43 total_revenue,  
44 previous_revenue,  
45 ROUND(  
46 (total_revenue - previous_revenue) / previous_revenue * 100,  
47 2  
48 ) AS growth_percent  
49 FROM revenue_with_previous  
50 ORDER BY month;
```

Data Output Messages Notifications

Showing rows: 1 to 4

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month	total_revenue	previous_revenue	growth_percent
timestamp without time zone	numeric	numeric	numeric
2007-02-01 00:00:00	8351.84	[null]	[null]
2007-03-01 00:00:00	23886.56	8351.84	186.00
2007-04-01 00:00:00	28559.46	23886.56	19.56
2007-05-01 00:00:00	514.18	28559.46	-98.20

Total rows: 4 Query complete 00:00:00.132 CRLF Ln 46, Col 45

	month timestamp without time zone	total_revenue numeric	previous_revenue numeric	growth_percent numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.00
3	2007-04-01 00:00:00	28559.46	23886.56	19.56
4	2007-05-01 00:00:00	514.18	28559.46	-98.20

Lab 3: Customer Lifetime Value (Running Total)

Query Query History

```
1 SELECT  
2 customer_id,  
3 payment_id,  
4 payment_date,  
5 amount,  
6 SUM(amount) OVER (  
7 PARTITION BY customer_id  
8 ORDER BY payment_date, payment_id  
9 ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW  
10 ) AS cumulative_spending  
11 FROM payment  
12 ORDER BY customer_id, payment_date, payment_id;  
13
```



```

25     c.last_name
26 )
27 SELECT
28     country,
29     customer_id,
30     first_name,
31     last_name,
32     total_spending,
33     spending_rank
34 FROM ranked_customer
35 WHERE spending_rank = 1
36 ORDER BY country;
37

```

	country character varying (50)	customer_id integer	first_name character varying (45)	last_name character varying (45)	total_spending numeric	spending_rank bigint
1	Afghanistan	218	Vera	Mccoy	67.82	1
2	Algeria	176	June	Carroll	151.68	1
3	American Samoa	320	Anthony	Schwab	47.85	1
4	Angola	528	Claude	Herzog	93.80	1
5	Anguilla	381	Bobby	Boudreau	99.68	1
6	Argentina	560	Jordan	Archuleta	129.71	1
7	Armenia	41	Stephanie	Mitchell	118.75	1
8	Austria	266	Nora	Herrera	113.74	1
9	Azerbaijan	334	Raymond	Mcwhorter	108.75	1
10	Bahrain	590	Seth	Hannon	108.76	1
11	Bangladesh	21	Michelle	Clark	146.68	1
12	Belarus	144	Clara	Shaw	189.60	1
13	Bolivia	431	Joel	Francisco	92.78	1
14	Brazil	178	Marion	Snyder	194.61	1
15	Brunei	91	Lois	Butler	107.66	1
16	Bulgaria	540	Tyrone	Asher	112.76	1
17	Cambodia	516	Elmer	Noe	92.76	1
18	Cameroon	352	Albert	Crouse	96.78	1
19	Canada	410	Curtis	Irby	167.62	1
20	Chad	535	Javier	Elrod	122.72	1
21	Chile	45	Janet	Phillips	124.74	1

Total rows: 108 Query complete 00:00:00.147